

CUSTOMER SEGMENTATION

Marketing Campaign Analysis

Unsupervised Learning Capstone Project
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EXECUTIVE SUMMARY

The Big Picture: Generic campaigns yield only 7% acceptance. By segmenting 2,208 customers into 3 groups using K-Means clustering, we can triple campaign effectiveness and unlock \$67K+ in incremental annual revenue.

3 Distinct Segments Identified

Premium Enthusiasts (25%), Moderate Mainstream (28%), Budget Families (47%) — validated by multiple algorithms

Income Drives Everything

Income is the #1 predictor of spending. Wine + Meat = 75% of all spend. Families with kids spend 14x less than childless households.

\$67K Revenue Opportunity

Targeted campaigns, channel optimization, and catalog cost savings can generate \$119K in benefits at \$50K setup cost (196% Year 1 ROI).

Key Next Steps

1. Deploy 3-tiered campaign strategy (Premium: catalogs, Mainstream: omnichannel, Budget: digital deals)
2. Reallocate 50% of budget to Premium segment (25% of customers, but highest ROI)
3. A/B test segment-specific vs. generic campaigns before full rollout

PROBLEM & SOLUTION SUMMARY

The Problem

Current state: All 2,208 customers receive identical campaigns across all channels.

Result: Average campaign acceptance of only ~7%.

Impact: Marketing spend wasted on unresponsive customers. High-value customers not receiving premium treatment. Budget customers getting irrelevant offers.

Root cause: No customer segmentation — one-size-fits-all approach.

The Solution

Approach: K-Means clustering on PCA-reduced behavioral data (10 components, 92% variance).

Why K-Means: Best silhouette score (0.29) among interpretable models. Confirmed by Hierarchical Clustering. Fast, scalable, deterministic.

Result: 3 distinct segments with clear behavioral profiles enabling targeted campaigns, channel optimization, and budget reallocation.

Business impact: **Projected \$67K incremental revenue. 196% ROI in Year 1.**

DATA & APPROACH OVERVIEW

2,208

Customers
Analyzed

17

Behavioral
Features

5

Algorithms
Compared

3

Segments
Identified

Key Data Insights

Higher income = higher spending

Income is the strongest predictor — customers earning \$77K spend 14x more than those earning \$35K

Wine & Meat dominate

These two categories represent 75% of total customer spending — the core revenue drivers

Children reduce spending

Families with kids spend \$99/yr vs. \$1,430/yr for childless households — fundamentally different buyer profiles

Campaigns underperform

Only 7% average acceptance rate — the one-size-fits-all approach wastes ~93% of campaign spend

Catalog = premium channel

High-income customers average 6.2 catalog purchases vs. 0.6 for low-income — a channel optimization opportunity

MODEL COMPARISON & SELECTION

Algorithm	Score	Clusters	Interpretability	Balance	Verdict
K-Means (k=3)	0.29	3	High	47/28/25%	SELECTED
K-Medoids (k=3)	0.27	3	High	Good	Backup
Hierarchical (Ward)	0.25	3	High	Good	Validates k=3
DBSCAN	0.47*	2-3	Low	Poor	Not suitable
GMM (k=3)	0.16	3	Medium	Good	Underperforms

**DBSCAN score inflated — excludes noise points from calculation, creates imbalanced unusable clusters.*

Why K-Means (k=3) Was Selected

- 1. Best usable silhouette score (0.29)
- 2. Independently confirmed by Hierarchical Clustering — two algorithms agree on 3 segments
- 3. Balanced clusters (47/28/25%) — each large enough for targeted campaigns
- 4. Directly interpretable: each segment maps to a clear business strategy
- 5. Production-ready: fast, scalable, new customers can be scored in milliseconds

THE THREE CUSTOMER SEGMENTS

BUDGET FAMILIES

46.9%

1,035 customers

Avg Income **\$35,204**

Avg Expenses/yr **\$99**

\$/Purchase **\$11.30**

Family Size **2.9**

Top Channel **Deals + Web**

Campaigns **0.17 accepted**

Price-sensitive. High web visits, low conversion. Respond only to discounts.

MODERATE MAINSTREAM

27.9%

615 customers

Avg Income **\$57,566**

Avg Expenses/yr **\$720**

\$/Purchase **\$33.80**

Family Size **2.8**

Top Channel **Omnichannel**

Campaigns **0.40 accepted**

Balanced spenders. Use all channels. Growth potential through cross-selling.

PREMIUM ENTHUSIASTS

25.3%

558 customers

Avg Income **\$76,658**

Avg Expenses/yr **\$1,430**

\$/Purchase **\$71.20**

Family Size **1.9**

Top Channel **Catalog + Store**

Campaigns **1.02 accepted**

High-income, small families. Wine + meat lovers. Most campaign-responsive.

RECOMMENDATIONS: CAMPAIGN STRATEGY

	Premium Enthusiasts	Moderate Mainstream	Budget Families
Campaign Type	Exclusive launches, VIP events	Cross-sell promotions, loyalty rewards	Deal alerts, family bundles
Channel	Catalog + In-store	Web + Store + Email	Mobile app + Email
Product Focus	Wine + Meat premium	Category expansion	Value packs, essentials
Budget Share	50% (\$50K)	30% (\$30K)	20% (\$20K)

Quick Wins

- ✓ Stop sending catalogs to Budget Families — saves \$37K/yr in print and postage costs (1,035 customers x \$3/piece x 12 months)
- ✓ Double catalog frequency for Premium — they average 6.2 catalog purchases and convert at the highest rate
- ✓ Launch mobile deal alerts for Budget — their 6.4 web visits/month shows digital engagement is already high

KEY ACTIONABLES FOR STAKEHOLDERS

Stakeholder	Action Required	Timeline	Priority
Executive Sponsor	Approve 3-tiered strategy + A/B test budget	Week 1	Critical
CRM Team	Tag all customers with segment labels	Week 1	Critical
Marketing Director	Design segment-specific campaign creative	Week 2-4	High
Data/Analytics	Deploy scoring model for new customers	Week 2-3	High
Channel Managers	Reallocate catalog vs. digital budget	Week 3-4	High

Implementation Roadmap

Phase 1 (Week 1-2)

CRM tagging + A/B test design + stakeholder approval

Phase 2 (Week 3-6)

A/B test: segment-specific vs. generic campaigns (500 customers per group)

Phase 3 (Week 7-8)

Analyze results, refine strategy, full rollout to all 2,208 customers

EXPECTED BENEFITS & COSTS

Annual Benefits (assumptions stated)

Benefit	Assumption	Value
Increased Premium campaign revenue	558 customers x lift from 1.02 to 1.50 campaigns x \$71.20/purchase	+\$41,000
Mainstream upselling	10% of 615 customers move to Premium behavior (\$720 to \$1,100 spend)	+\$26,000
Catalog cost savings	Stop catalogs to 1,035 Budget customers x \$3/piece x 12 months	+\$37,000
Reduced wasted ad spend	Reallocating ineffective generic campaigns	+\$15,000
TOTAL ANNUAL BENEFITS		\$119,000

Costs

Cost Item	One-Time	Annual
Campaign redesign (3 tracks)	\$15,000	—
CRM/data integration + mobile app	\$30,000	\$10,000
A/B testing + additional catalogs	\$5,000	\$11,000
TOTAL COSTS	\$50,000	\$21,000

Year 1 ROI

196%

\$98K net benefit / \$50K investment

RISKS, CHALLENGES & FURTHER ANALYSIS

Risk	Likelihood	Impact	Mitigation
Data staleness (2012-14 data)	High	High	Re-run on current data before launch
Segment migration over time	Medium	Medium	Quarterly re-scoring; monitor drift
Budget reallocation resistance	Medium	High	A/B test results prove ROI; phase gradually
Over-personalization fatigue	Low	Medium	Maintain opt-out; vary messaging within segments

Further Analysis Required

Customer Lifetime Value (CLV)

Add time dimension to understand long-term segment value beyond current spending

Churn Prediction per Segment

Identify at-risk Premium customers and prioritize retention efforts

Geographic Segmentation

No location data available — adding this enables regional campaign optimization

A/B Testing Validation

Run controlled experiments before full rollout to validate projected lift

CONCLUSION

3 Actionable Segments

K-Means clustering identified Premium Enthusiasts (25%), Moderate Mainstream (28%), and Budget Families (47%) — each with distinct income, spending, channel, and campaign behavior.

\$98K Net Year 1 Impact

Through targeted campaigns, channel optimization, and catalog cost savings — at a 196% ROI on \$50K investment.

Clear Implementation Path

8-week phased rollout with CRM tagging, A/B testing, and full deployment. Each stakeholder has a defined action and timeline.

Recommendation: Approve the 3-tiered campaign strategy and begin Phase 1 (CRM tagging + A/B test design) immediately.

Thank you